Day 1: SAT/TSI Challenge – Exponents & Radicals

1. **SAT:** If $4^x = 8$, what is x?

   A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. 2 D. $\frac{4}{3}$

2. **SAT:** The expression $\dfrac{(2x^3y^{-2})^3}{4x^{-3}y^4}$ is equivalent to:

   A. $\dfrac{2x^{12}}{y^{10}}$ B. $\dfrac{x^{12}}{2y^{10}}$ C. $2x^6y^{-2}$ D. $\dfrac{2x^6}{y^{10}}$

3. **TSI:** If $a^x = 3$ and $a^y = 5$, what is a^{2x-y}?

   Answer: ______________

4. **SAT:** Simplify completely: $\sqrt[4]{16x^{12}y^8} \cdot \sqrt{x^3y^2} =$ ____________________

Day 2: SAT/TSI Challenge – Polynomials & Division

1. **SAT:** If $p(x) = x^3 - kx^2 + 2x + 8$ and $(x+2)$ is a factor of $p(x)$, what is k?

   A. -3 B. -1 C. 1 D. 3

2. **TSI:** Factor completely: $x^4 - 16 =$ __________________________________

   (Hint: Difference of squares twice, or recognize $x^4 - 16 = (x^2)^2 - 4^2$)

3. **SAT:** The polynomial $p(x)$ has roots at $x = -2$, $x = 1$, and $x = 3$. If $p(x)$ has a leading coefficient of 2, what is $p(0)$?

   A. -12 B. -6 C. 6 D. 12

4. **TSI:** When $2x^3 - 5x^2 + 3x - 7$ is divided by $(x-2)$, what is the remainder?

   Remainder: ______________

Day 3: Final SAT/TSI Mixed Challenge

1. **SAT:** Factor completely: $8x^3 - 27y^3 =$ __________________________________

2. **SAT:** If $f(x) = x^3 + ax^2 + bx - 6$ has factors $(x-1)$ and $(x+2)$, what is $a+b$?

   A. -4 B. -2 C. 2 D. 4

3. **TSI:** Which expression is equivalent to $\dfrac{\sqrt[3]{x^5}}{\sqrt{x}}$?

   A. $x^{7/6}$ B. $x^{5/6}$ C. $x^{1/6}$ D. $x^{13/6}$

4. **SAT:** The graph of $y = f(x)$ has x-intercepts at $(-3, 0)$, $(0, 0)$, and $(2, 0)$. Which could be $f(x)$?

   A. $x(x+3)(x-2)$ B. $x(x-3)(x+2)$ C. $(x+\ \ 3)(x-2)$ D. $x^2(x+3)(x-2)$

5. **Challenge:** If $x^3 + y^3 = 91$ and $x + y = 7$, find xy.

   Hint: Use $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$ and $(x+y)^2 = x^2 + 2xy + y^2$

Answer: $xy =$ ___________

Key Strategies: *(1) When stuck, try plugging in answer choices. (2) Factor before simplifying when possible. (3) Remember: if r is a root, then $(x - r)$ is a factor. (4) Check your work by expanding!*

Answer Key: D1: 1. B 2. A 3. 9/5 or 1.8 4. $2x^{9/2}y^3$ or $2x^4y^3\sqrt{x}$ **D2:** 1. A 2. $(x-2)(x+2)(x^2+4)$ 3. D 4. R=1 **D3:** 1. $(2x-3y)(4x^2+6xy+9y^2)$ 2. B 3. A 4. A 5. $xy = 10$